

Trustworthy Machine Learning, 2026

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Privacy: Membership Inference Attack

Goal

Participants are given a trained image classification model and a dataset of samples.

For each sample, participants must predict whether it was part of the model's training dataset.

Each submission must assign a score between 0 and 1 for every sample:

- A higher score indicates a higher likelihood that the sample is a member (used during training).
- A lower score indicates a higher likelihood that the sample is a non-member.

Challenge

The dataset is constructed such that:

- Members and non-members are drawn from the same underlying distribution
- No explicit indicator of membership is provided

A successful attack must generalize across all samples and produce a strong ranking between members and non-members.

Evaluation Metric

Submissions are evaluated using:

$$Score = TPR@5\%FPR$$

Where:

- TPR (True Positive Rate): fraction of members correctly identified
- FPR (False Positive Rate): fraction of non-members incorrectly identified

Higher values indicate better membership inference performance.

Dataset

Public Dataset (pub.pt): Contains:

- id
- image
- label
- membership (1 = member, 0 = non-member)

This dataset is provided for analysis, debugging and developing attack strategies

Private Dataset (priv.pt): Contains:

- id
- image
- label
- membership = None

Participants must predict membership for this dataset. This is the dataset used for evaluation. Contains 14k samples.

Model

Participants are provided with a pretrained model:

- Architecture: ResNet-18
- Trained on a subset of the dataset (members only)

Dataset Access

You can obtain the dataset and template as follows:

```
wget "https://huggingface.co/datasets/SprintML/tml26_task1/resolve/main/pub.pt" \
"https://huggingface.co/datasets/SprintML/tml26_task1/resolve/main/priv.pt" \
"https://huggingface.co/datasets/SprintML/tml26_task1/resolve/main/model.pt" \
"https://huggingface.co/datasets/SprintML/tml26_task1/resolve/main/task_template.py"
```

Additional Resources

You are supposed to submit your scores using the code file `task_template.py` provided.

The template includes example code to load the model, load the datasets (`pub.pt` and `priv.pt`), generate a random submission and submit results to the server.

Remember to replace `YOUR_API_KEY_HERE` with your actual API-key (keep the double quotes), and replace `OUTPUT_FILE` path with your real results.

Submissions must follow the required file format described below.

Submission Format

The submission file must be named `submission.csv` with an `id` and a `score` field:

```
id,score
123,0.82
124,0.11
```

Requirements

- Each sample must appear exactly once
- Score must be a numeric value
- Score must lie in the range $[0, 1]$
- No duplicate or missing IDs
- File must be a valid CSV
- Submissions that violate these requirements will be rejected automatically.

Submissions that do not meet these requirements will be rejected automatically.

Scoring

All submissions are automatically validated and evaluated on the hidden test set.

- **Public leaderboard:** based on 30% of `priv.pt` samples
- **Private leaderboard:** final ranking based on the remaining 70%

The leaderboard score reflects the `score` achieved by the submission. Higher is better.

Leaderboard

Results are published at: http://34.63.153.158/leaderboard_page

The leaderboard shows the best result per team only. If your current submission score is lower than the score saved in the leaderboard, the score will not be updated.

Points to note regarding submission:

You can only submit a CSV with scores **every 60 minutes**. If your submission is malformed or wrong, you should get a comprehensive error message. However, even if you fail, you get a **2 minute** cooldown.

Read the following instructions carefully:

Additionally, you are required to submit a **Report PDF file** to CMS that explains your approach to the task. This file should contain:

- An Introduction:
 - A short paragraph explaining **your understanding of the task** written by your own words.
- A Main body:
 - Explaining **your approach to the task** including the details about the hyperparameters and structural choices you have made, as well as the reasons you made that choice.
 - Explaining **key results** that you have obtained on the public leaderboard.
- A Conclusion
 - In your conclusion try to be goal oriented. In other words, try to answer this question: now that we have a successful MIA method, what practical implications does it have and why should we be concerned about it?

The report should NOT be more than **2 pages** long. Also, make sure you include the URL to the Github repository that contains your code. This repository should contain a readme file that ONLY explains how to **recreate your best result** that appears on the leaderboard. The reports that do not contain the URL to github repo will NOT be considered for evaluation. For the report you SHOULD use the [template for ICLR](#) on overleaf. Note that there is no need for an abstract and you should skip it.

The submission to the leaderboard will end automatically on **06.05.2026 00:00** and after that you will not be able to submit any results to the leaderboard. The same date also applies to CMS. As a result, make sure to submit your results and report on time. Thereafter, you will be able to check your evaluation results performed on the held-out set on the leaderboard.

References

1. Membership Inference Attacks Against Machine Learning Models; Shokri et al.: <https://arxiv.org/pdf/1610.05820>
2. Membership Inference Attacks From First Principles; Carlini et al.: <https://arxiv.org/pdf/2112.03570>
3. Low-Cost High-Power Membership Inference Attacks; Zarifzadeh et al.: <https://arxiv.org/pdf/2312.03262>